Three counterexamples to the Akbari–Cameron–Khosrovshahi row-space conjecture

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Abstract

We give three connected simple graphs on 52 vertices whose adjacency matrices have real rank 14 and whose only Boolean row-space vectors are zero and the adjacency rows. Their different edge counts make them mutually nonisomorphic. Each graph refutes the universal Akbari–Cameron–Khosrovshahi adjacency-row-space conjecture. For all three graphs, Lean 4 verifies the simple-adjacency conditions, the existence of an edge, the classification of all Boolean vectors in the real row space, and the resulting negation of the conjecture. The finite classification checks all $2^{14} = 16,384$ Boolean prefixes using exact integer certificates. We also give a mathematical blowup argument producing connected counterexamples of every order at least 52.

Statement and result

For a real matrix A, let $\mathrm{Row}_{\mathbb{R}}(A)$ denote its real row space. A Boolean vector has every coordinate in $\{0, 1\}$.

The universal ACK adjacency-row-space conjecture asserts that every finite simple graph with at least one edge has a nonzero Boolean vector in the real row space of its adjacency matrix that is not an adjacency row. The originating Question 2 of Akbari, Cameron, and Khosrovshahi [1] asks which graphs have this property. The universal assertion is explicitly stated as Conjecture 1.1 by Akansha and Sivakumar [3]. Sciriha et al. [2] study restrictions on potential counterexamples.

Theorem 1. *For each of the explicitly specified graphs $G_{550}, G_{563}, G_{570}$ below,*

$$\mathrm{Row}_{\mathbb{R}}(A_G) \cap \{0,1\}^{52} = \{\mathbf{0}\} \cup \{\textit{rows of } A_G\}.$$

Each graph is connected, simple, and reduced, and has rank 14. In particular, the ACK universal row-space conjecture is false.

Graph	Vertices	Edges	Real rank	Boolean vectors
G_{550}	52	550	14	53
G_{563}	52	563	14	53
G_{570}	52	570	14	53

The Boolean-vector count includes zero. All 52 adjacency rows are distinct and nonzero. The different edge counts prove the graphs are mutually nonisomorphic.

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Scope

The three examples refute the same universal adjacency-row-space assertion. They do not settle the separate maximum-order or extremal-classification conjectures for reduced graphs of a given rank. We make no minimality claim for order 52 or rank 14.

The formal proof sources are available at

https://github.com/alexgocardinal/ack-rowspace-counterexample.

The verified proof revision is 5b04adb. The repository is currently private; access is required.

Exact construction

All mask coordinates below are indexed $0, \ldots, 13$: coordinate j is bit j, so the least-significant bit is coordinate 0. Let b be the following list of 14 masks:

```
4018, 11465, 5136, 6034, 8429, 721, 3634, 7995,
14985, 1513, 6863, 13763, 3468, 2322
```

Define the 14×14 matrix B by $B_{ij} = \mathrm{bit}_j(b_i)$. It is symmetric with zero diagonal, and $\det(B) = -9$. For each graph take its following list c of 38 masks, and let W be the 52×14 Boolean matrix whose row masks are the concatenation (b, c). Define

$$A = WB^{-1}W^T.$$

This formula, involving only the integers displayed here, uniquely specifies each graph. Exact arithmetic verifies that every entry of A is 0 or 1 and every diagonal entry is zero. The first 14 rows and columns of A form B.

G_{550}.

```
384, 676, 1497, 1517, 1810, 2031, 2444, 2692,
2859, 2998, 3080, 3362, 3464, 4289, 4559, 4803,
5378, 5520, 5650, 5715, 6099, 6162, 6227, 6284,
6349, 6566, 8385, 9232, 9248, 9904, 10761, 12091,
13379, 13807, 14035, 14051, 14635, 15410
```

G_{563}.

```
384, 449, 793, 1129, 1296, 1452, 1810, 2031,
2060, 2692, 3080, 3390, 3451, 4875, 5378, 5520,
6099, 6566, 6611, 6665, 6838, 7586, 8385, 8745,
9232, 9616, 12739, 13698, 13807, 13881, 14107, 14123,
14563, 15488, 963, 5650, 6349, 10445
```

G_{570}.

```
89, 384, 493, 963, 1497, 2617, 2843, 3120,
3346, 3362, 3712, 4112, 5152, 5241, 5467, 6099,
6182, 6271, 6284, 8385, 8457, 9011, 9089, 9248,
9632, 10278, 10445, 10662, 12169, 12355, 13011, 13537,
13609, 13807, 14051, 14883, 14989, 15488
```

The repository supplies the full adjacency matrices as lists of integer row masks in the Lean witness files. In particular, the formulas specify ordinary simple unweighted graphs; the intermediate rational inverse does not define weighted edges.

Proof and exact certificates

Write $W = \binom{B}{U}$, with U a 38×14 Boolean matrix. Then

$$A = \begin{pmatrix} B & U^T \\ U & UB^{-1}U^T \end{pmatrix}.$$

Its Schur complement is zero, so $\operatorname{rank}_{\mathbb{R}}(A) = 14$. More explicitly, every row-space vector has the unique form

$$f(x) = (x,\ xB^{-1}U^T), \qquad x \in \mathbb{R}^{14}.$$

Indeed, the first 14 rows of A span its row space, and their nonsingular first block is B. A Boolean row-space vector must therefore have $x \in \{0,1\}^{14}$. There are exactly $2^{14} = 16,384$ assignments to check.

For each graph, exact rational arithmetic evaluates $xB^{-1}U^T$ for all these assignments. Exactly 53 assignments produce a Boolean vector. Their prefixes are zero and the 52 row masks of W; their full vectors are zero and the adjacency rows of A. The remaining 16,331 assignments each have a coordinate outside $\{0,1\}$. This proves the theorem.

The displayed construction also permits direct checks of the edge counts, connectivity, and distinctness of the adjacency rows. The nonsingularity of B and the block identity above prove the rank assertion without a floating-point rank computation.

Lean formalization

The repository formalizes the conjecture over $\mathbb{R}$. Its predicate `InRealRowSpace` explicitly requires a vector to be a real linear combination of adjacency rows. The predicate `IsSimpleAdjacency` requires zero diagonal, symmetry, and entries in $\{0,1\}$. The conjecture also explicitly requires an edge and excludes both the zero vector and every existing adjacency row.

For each graph, an integer matrix D is checked to satisfy

$$3A_{ij} = \sum_{k=0}^{13} A_{ik}D_{kj}.$$

Linearity then gives the same reconstruction for every vector in the real row space. If its first 14 coordinates have mask m, its reconstructed coordinate j is

$$\frac{1}{3} \sum_{\substack{0 \le k < 14 \\ \mathrm{bit}_k(m)=1}} D_{kj}.$$

A graph-specific finite certificate assigns each mask either to zero, to an adjacency-row prefix, or to a coordinate whose integer numerator is neither 0 nor 3. Lean checks every certificate entry. A structural proof of binary encoding ensures that every Boolean prefix is represented, and reconstruction uniqueness identifies the accepted vectors.

The final counterexample declarations are:

```
ACK.Witness550.isCounterexample
ACK.Witness563.isCounterexample
ACK.Witness570.isCounterexample
```

Each namespace also contains `conjecture_false`, proving the negation of `ACK.Conjecture`. The axiom reports for all six declarations contain exactly `propext`, `Classical.choice`, and `Quot.sound`. They contain neither `sorryAx` nor `Lean.ofReduceBool`. The formal verification requires no external SAT solver or Python execution.

The current formalization proves the counterexample property and the refutation. The rank argument, the additional graph invariants, and the blowup corollary are not separately formalized.

An infinite family

Replace each vertex i by a nonempty independent set of size t_i and each edge by the complete bipartite graph between its endpoint sets. Let T be the resulting membership matrix, so $A' = TAT^T$. Since every set is nonempty, T has a left inverse L. Thus $AT^T = LA'$, while $A' = T(AT^T)$, so these two matrices have the same row space. Consequently,

$$\mathrm{Row}(A') = \{xT^T : x \in \mathrm{Row}(A)\}.$$

The vector xT^T is Boolean if and only if x is Boolean. Every nonzero such x is a row of A, and its repeated-coordinate image is a row of A'. Hence every blowup is again an ACK counterexample, with rank 14. Taking one t_i arbitrary and all others 1 yields a connected counterexample of every order at least 52. This is a corollary of the same refutation, not a separate conjecture settlement.

Reproduction

The repository contains the explicit graphs, all integer reconstruction and classification certificates as Lean source, and the complete formal proofs. The toolchain is Lean 4.35.0, with Mathlib pinned to revision `56a005855ea17e0b0958f73ec801ae140b0a1679`. The dependency manifest is committed.

After installing Lean through elan and obtaining repository access, run the following commands from its root:

```
lake exe cache get
lake build
lake env lean Verification.lean
```

The final command displays the formal statement, the three counterexample theorem types, and the axiom dependencies of all three counterexample and refutation theorems. The GitHub workflow also builds the project and runs this verification entry point.

References

[1] S. Akbari, P. J. Cameron, G. B. Khosrovshahi. *Ranks and signatures of adjacency matrices*. Preprint (2004), Question 2. https://webspace.maths.qmul.ac.uk/p.j.cameron/preprints/ranksign.pdf

[2] I. Sciriha, B. Kaur, J. L. Borg, M. Debono. *Potential counter-examples to a conjecture on the column space of the adjacency matrix*. Special Matrices 13 (2025), 20250033. https://doi.org/10.1515/spma-2025-0033.

[3] S. Akansha, K. C. Sivakumar. *Affirmative Results on a Conjecture on the Column Space of the Adjacency Matrix*. arXiv:2601.02746v1, 6 January 2026, Conjecture 1.1 and reference [6]. https://arxiv.org/abs/2601.02746